Quantum Computing Workshop

Student Lab Handout

Session Overview

This 75-minute lab focuses on hands-on matrix transformations in quantum computing. You will calculate, verify, predict, and analyze quantum circuits, bridging theory with simulation results through histograms and custom design.

Session Objectives

- Calculate and verify matrix transformations for quantum operations
- Predict quantum states before running simulations
- Match mathematical theory to experimental histograms
- Design and analyze custom quantum circuits
- Submit complete calculations with screenshots for assessment

Timed Agenda (75 minutes)

Time	Activity
0:00-0:10	**Introduction & Setup (10 min)** Review lab goals and quantum matrix basics Setup simulation environment and test basic circuits
0:10-0:25	**Core Calculations (15 min)** Calculate matrix transformations (individual work) Verify results using provided checker
0:25-0:40	**State Prediction & Simulation (15 min)** Predict states mathematically Run simulations and capture histograms
0:40-0:55	**Theory-to-Histogram Matching (15 min)** Compare predicted vs. simulated results Analyze discrepancies and key patterns
0:55-1:10	**Custom Circuit Design (15 min)** Design personal circuit using learned transformations Simulate and analyze outcomes
1:10-1:20	**Documentation Review (10 min)** Compile calculations + screenshots Peer review format and completeness
1:20-1:25	**Submission & Q&A (5 min)** Final submission instructions Address questions and wrap-up

State & Matrix Reference (Use Throughout)

Basis States:

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \qquad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \qquad |01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}$$

$$|10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \qquad |11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}$$

Bell States:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix}$$

$$\left|\Phi^-\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle - |11\rangle\right) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\-1\end{pmatrix}$$

$$\left|\Psi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\\0\end{pmatrix}$$

$$\left|\Psi^-\right\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right) = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\\0\end{pmatrix}$$

$$\text{Hadamard: } H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix}$$

$$\text{X (NOT): } X = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$$

$$\text{Pauli-X: } X = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$$

$$\text{Pauli-Y: } Y = \begin{bmatrix}0 & -i\\i & 0\end{bmatrix}$$

$$\text{Pauli-Z: } Z = \begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$$

$$\text{S-Gate: } S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

$$\text{Rx}(\theta)\text{: } \text{Rx}(\theta) = \begin{bmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

$$\text{Ry}(\theta)\text{: } \text{Ry}(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

$$\text{Rz}(\theta)\text{: } \text{Rz}(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

$$\text{CNOT } (q_0 \to q_1)\text{: } \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Setup (5 min)

```
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
from qiskit.visualization import plot_histogram
simulator = AerSimulator()
```

Part 1: Superposition - Complete Matrix Math (15 min)

Challenge 1.1: $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Calculate step-by-step:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \xrightarrow{\text{Hadamard}} H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} ? \\ ? \end{pmatrix} = |+\rangle$$

Calculation: $|\alpha|^2 = ?$, $|\beta|^2 = ?$ (probabilities)

Verify with Code:

```
qc_super = QuantumCircuit(1,1)
qc_super.h(0)
qc_super.measure(0,0)
job = simulator.run([qc_super], shots=1024)
plot_histogram(job.result().get_counts(0))
```

Question: Histogram matches your $|\alpha|^2$, $|\beta|^2$? _____

Part 2: Bell State - Step-by-Step Matrix (20 min)

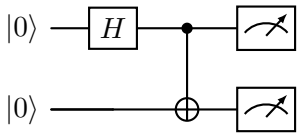
Initial State: $|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

Step 1: Apply $H \otimes I$ (H on q_0, Identity on q_1):

$$(H \otimes I)|00\rangle = ?$$

Step 2: Apply CNOT:

$$\text{CNOT}? = ?$$



Code + Verify:

```
bell = QuantumCircuit(2,2)
bell.h(0)
bell.cx(0,1)
bell.measure([0,1],[0,1])
job = simulator.run([bell], shots=1024)
plot_histogram(job.result().get_counts(0))
```

Correlation: $\frac{\text{counts}(00)+\text{counts}(11)}{\text{total}} =?$ _____

Part 3: Modified Bell - Predict + Calculate (15 min)

Challenge: Add X to qubit 1 BEFORE CNOT.

Calculate new final state:

$$|00\rangle \xrightarrow{I\otimes X} |01\rangle \xrightarrow{H\otimes I} ? \xrightarrow{\text{CNOT}} ?$$

Prediction: Which states in histogram? _____

```
qc_mod = QuantumCircuit(2,2)
qc_mod.x(1)      # NEW: Flip q1 to |1>
qc_mod.h(0)      # H on q0
qc_mod.cx(0,1)   # CNOT
qc_mod.measure_all()
job = simulator.run([qc_mod], shots=1024)
plot_histogram(job.result().get_counts(0))
```

Results: Matches prediction? Explain: _____

Part 4: Design Challenge - Matrix Engineering (10 min)

Team Mission: Create circuit with **75%** $|00\rangle$, **25%** $|11\rangle$.

Hint: Use different superposition weights before CNOT, for instance, by replacing standard Hadamard (50/50) with a biased rotation.

```
qc_design = QuantumCircuit(2,2)
# Team solution: modify superposition weights
qc_design.h(0)  # Start here, then adjust
qc_design.cx(0,1)
qc_design.measure_all()
```

Your math prediction: Final state coefficients? _____

Lab Report (Submit ALL)

1. **3 Screenshots:** Superposition, Bell, Modified Bell histograms
2. **Full calculations:**
   - $H|0\rangle =?$ (Part 1)
   - Bell step-by-step (Part 2)
   - Modified Bell prediction (Part 3)
3. **Design challenge:** Code + math + histogram
4. **Correlation:** Bell (Part 2) = _____

Success: Math predictions match histograms within 5%!